Horizontal & Vertical Components of Projectiles

Exam code: 9MA0

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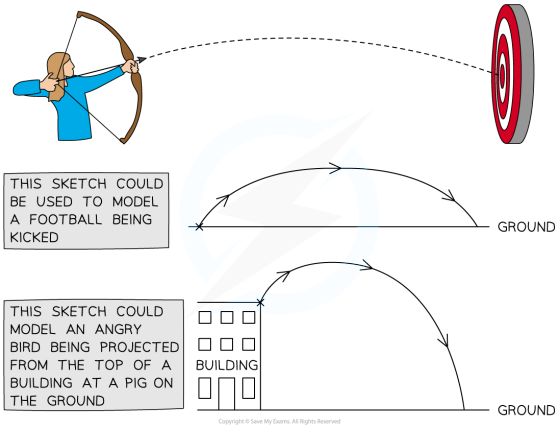
Horizontal & vertical components

Help me understand

What is a projectile?

- A projectile is a particle moving freely, under gravity, in a two-dimensional plane
- Examples of projectile motion include
  - Sports such as basketball, javelin, archery, etc
  - Computer games such as Angry Birds



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Ask questions and get help with exactly what you're revising.
Got it

- The modelling assumptions with projectiles are
  - **No air resistance**, no horizontal forces
  - The projectile **moves freely** under gravity (no other vertical forces)
  - **Ignore spin** and the **rotation** of the projectile
  - Motion is **symmetrical** along the path of a **parabola**

What is the acceleration of a projectile?

- As it is 2D there will be a horizontal component to acceleration (**a_x**)and a vertical component (**a_y**)
- There will be **no** horizontal acceleration (**$a_x = 0$**)
  - No forces are acting horizontally
  - **Horizontal speed** is **constant**

- There will be constant vertical acceleration (due to gravity, **$a_y = \pm g$**)
- **a_y** may be positive or negative depending on which direction is taken as positive
- The acceleration can be written as 2D vector

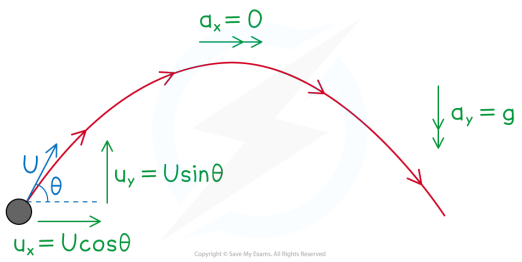
$$a = 0i - gj = \begin{pmatrix} 0 \\ -g \end{pmatrix} m\,s^{-2}$$

What is the initial velocity of a projectile?

- As it is 2D there will be a horizontal component to the **initial velocity**(**u_x**) and a vertical component (**u_y**)
- A projectile is launched with **initial speed** U m s^{-1} at an angle θ to the horizontal
  - If it is projected **below** the horizontal then θ would be **negative**

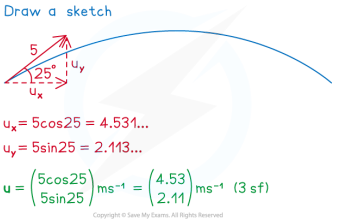
- Its initial velocity, **u** m s^{-1}, is a vector with:
  - horizontal component, **u_x** = U cosθ
  - vertical component, **u_y** = U sinθ

- The initial velocity can have a **positive** or **negative vertical component** as an object can be projected **upwards** or **downwards**



Worked Example

A projectile is launched at an angle of 25° to the horizontal with speed 5 m s^{-1} .



Examiner Tips and Tricks

- Sometimes they may give you the initial velocity as a vector already. To find the initial speed you would find the magnitude of the initial velocity and to find the angle of projection you could sketch a diagram and use trigonometry.
- The horizontal speed is always constant so there will never be a point during the motion when the speed of the object is zero. Its minimum speed will be at its maximum height when its vertical velocity is instantaneously zero.

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